

📄 EXAM FORMULA SHEET • LAST-DAY REVISION

Basic Electronic Circuits & Instrumentation

UNIT II • BEEE • B.TECH 1ST YEAR

A compact, exam-ready revision weapon covering Rectifiers, Power Supplies, Amplifiers, and Electronic Instrumentation — with all key formulas, tables, flashcards, and exam tips.

SUBJECT	UNIT	SEMESTER	LEVEL
BEEE	UNIT II	1st SEM	B.TECH

§1 DC Power Supply Block Diagram

§2 Rectifiers & Key Formulas

§3 Capacitor Filter & Zener Regulator

§4 CE Amplifier & PA System

§5 Frequency Response & Bandwidth

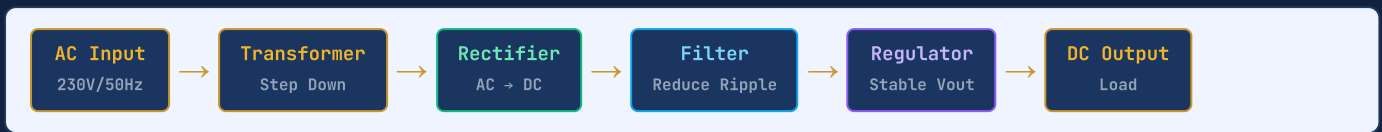
§6 Electronic Instrumentation

§7 Comparison Tables

§8 Flashcards (25 Q&A)

§9 Exam Tips

§10 Quick Revision Sheet



Block	Function	Key Device
Transformer	Steps down 230V AC to lower voltage; provides electrical isolation	Iron-core transformer
Rectifier	Converts AC → pulsating DC (uni-directional flow)	PN Junction Diode(s)
Filter	Smooths pulsating DC by reducing AC ripple component	Capacitor / Inductor
Regulator	Maintains constant output voltage despite load/input variation	Zener Diode / IC 7805

Circuit

- ▶ **+ve cycle:** D1, D3 conduct; D2, D4 OFF
- ▶ **-ve cycle:** D2, D4 conduct; D1, D3 OFF
- ▶ Both half cycles used → output at 100 Hz
- ▶ No center-tap required; preferred in practice

HWR OUTPUT (pulsating DC):

FWBR – AVERAGE OUTPUT VOLTAGE

$$V_{dc} = 2V_m / \pi \approx 0.636 V_m$$

Twice that of HWR; both halves used

RIPPLE FACTOR (Γ)

$$Y = V_{ac} / V_{dc} = \sqrt{[(V_{rms}/V_{dc})^2 - 1]}$$

HWR: $\gamma = 1.21$ | FWBR: $\gamma = 0.48$

FORM FACTOR (FF)

$$FF = V_{rms} / V_{dc}$$

HWR: FF = 1.57 | FWBR: FF = 1.11



- ▶ Reduces ripple (AC component) in rectified output
- ▶ Provides smoother DC to the load
- ▶ Capacitor placed in **parallel** with load R_L

- 1 **Charging Phase:** Diode ON → capacitor charges to peak voltage V_m
- 2 **Discharging Phase:** Diode OFF → capacitor slowly discharges through R_L , maintaining current flow
- 3 This charge/discharge cycle reduces voltage dips → smoother output
- 4 Larger capacitance → smaller ripple → better filtering

RIPPLE VOLTAGE (APPROXIMATE)

$$V_r = I_{dc} / (f \cdot C)$$

f = frequency; C = capacitance (F); I_{dc} = average current

RIPPLE FACTOR WITH FILTER

$$\gamma = 1 / (4\sqrt{3} \cdot f \cdot C \cdot R_L)$$

Smaller γ = better filter performance

WITHOUT FILTER (pulsating):

WITH CAPACITOR FILTER:



Zener Breakdown

$V_{in} \rightarrow [R_s \text{ (series resistor)}] \rightarrow \text{---} V_{out} \text{ (to } R_L)$

$\quad \quad \quad \perp \text{ [Zener } D_Z] \rightarrow \text{GND}$

- 1 Unregulated DC applied across circuit with series R_s
- 2 Zener conducts in reverse breakdown at V_Z
- 3 Excess voltage dropped across R_s
- 4 Output across Zener stays constant $= V_Z$

SERIES RESISTOR CURRENT

$$I_s = (V_{in} - V_Z) / R_s$$

Total current through series resistor

CURRENT DISTRIBUTION

$$I_S = I_Z + I_L$$

I_Z = zener current; I_L = load current

LOAD VOLTAGE (REGULATED)

$$V_{out} = V_Z = \text{constant}$$

Output stays constant despite V_{in} or R_L changes

- ▶ Voltage reference in power supplies
- ▶ Overvoltage protection
- ▶ Clipping / clamping circuits
- ▶ Meter protection



Parameter	Half Wave Rectifier (HWR)	Full Wave Bridge Rectifier (FWBR)
No. of Diodes	1	4
Output Frequency	f (50 Hz)	2f (100 Hz)
V _{dc}	0.318 V _m	0.636 V _m
Ripple Factor	1.21 (high)	0.48 (low)

Efficiency	40.6%	81.2%
Center Tap Required?	No	No (Bridge) / Yes (CT type)
Transformer Utilization	Poor (0.287)	Better (0.693)
Peak Inverse Voltage	V_m	V_m
Applications	Low-power, signal detectors	Most electronic power supplies



§ 3.1



STAGE	DEVICE	FUNCTION
Microphone (Transducer)	Condenser / Dynamic mic	Converts sound waves → electrical signal (mV range)
Pre-Amplifier	BJT / Op-Amp	Amplifies weak microphone signal for further processing
Power Amplifier	Class AB BJT stage	Provides sufficient power to drive the loudspeaker
Loudspeaker	Electromagnetic speaker	Converts amplified electrical signal back to sound waves



§ 3.2

CIRCUIT COMPONENTS

NPN Transistor in CE Configuration

Input: Base–Emitter junction (V_{BE})

Output: Collector–Emitter junction (V_{CE})

COMPONENT	ROLE
R_1, R_2	Voltage divider biasing (DC operating point)
R_C	Collector resistor — provides voltage gain
R_E	Emitter resistor — DC stability
C_1, C_2	Coupling capacitors — block DC, pass AC
C_E	Bypass capacitor — shorts R_E for AC, max gain
V_{CC}	Supply voltage

WORKING PRINCIPLE

- 1 Small AC input signal applied at base via coupling cap C_1
- 2 Signal modulates base current I_B
- 3 Transistor amplifies: $I_C = \beta \cdot I_B$
- 4 Amplified I_C flows through $R_C \rightarrow$ creates output voltage
- 5 Output taken at collector via coupling cap C_2
- 6 **Phase reversal (180°)**: +ve input $\rightarrow$ -ve output

⚠ KEY CHARACTERISTIC

Phase shift = 180°

Input and output are out of phase in CE configuration

CE AMPLIFIER — GAIN FORMULAS

VOLTAGE GAIN

$$A_V = -\beta \cdot R_C / r_e$$

$r_e = 26\text{mV}/I_E$ (dynamic emitter resistance)
-ve sign = 180° phase shift

CURRENT GAIN

$$A_i = I_C / I_B = \beta = h_{fe}$$

Also written as h_{FE} (DC) or h_{fe} (AC)
Typical: $\beta = 50$ to 300

POWER GAIN

$$A_P = A_V \times A_i$$

In dB: $A_P(\text{dB}) = 10 \log(P_{out}/P_{in})$

GAIN IN DB

$$A_V(\text{dB}) = 20 \log|A_V|$$

Voltage & current gains use $20 \log$

SIMPLIFIED VOLTAGE GAIN

$$A_V \approx -R_C / R_E$$

Without bypass cap (approximate)

THREE FREQUENCY REGIONS

REGION	FREQUENCY	GAIN BEHAVIOR	CAUSE
Low Freq	$f < f_L$	Falls ↓	Coupling / bypass caps act as open circuits → reduced gain
Mid Freq	$f_L < f < f_H$	Flat / Maximum	All caps act ideally → maximum stable gain
High Freq	$f > f_H$	Falls ↓	Internal junction capacitances → gain reduction

BANDWIDTH (BW)

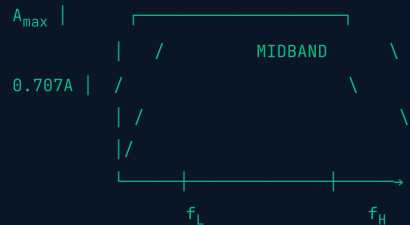
$$BW = f_H - f_L$$

f_L = lower cutoff freq | f_H = upper cutoff freq

At cutoff: Gain = $A_{max}/\sqrt{2} = 0.707 \times A_{max}$ (−3 dB point)

FREQUENCY RESPONSE GRAPH (TEXT)

Gain (dB) vs Frequency (log scale)



−3 dB Points

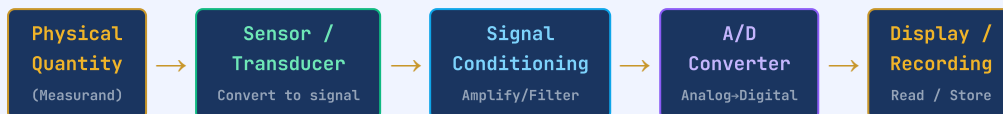
The frequencies f_L and f_H where gain drops to 70.7% of maximum — these define the usable bandwidth of the amplifier.

PARAMETER	CE AMPLIFIER VALUE / NATURE	REMARKS
Voltage Gain (A_v)	High (50–1000)	Most commonly used for voltage amplification
Current Gain (A_i)	β (50–300)	Equal to transistor beta
Power Gain	Very High	$A_p = A_v \times A_i$
Phase Shift	180° (inverted)	Output is out of phase with input
Input Impedance	Medium (1–5 k Ω)	Moderate loading on signal source
Output Impedance	Medium-High (40 k Ω)	Not suitable for low-impedance loads directly
Applications	Audio amplifiers, radio receivers, general signal amplification	Most versatile BJT configuration

PARAMETER	COMMON EMITTER (CE)	COMMON BASE (CB)	COMMON COLLECTOR (CC)
Voltage Gain	High	High	<1 (Unity)
Current Gain	High (β)	<1 (α)	High ($\beta+1$)
Power Gain	Highest	Medium	Medium
Phase Shift	180°	0° (no reversal)	0° (no reversal)
Input Impedance	Medium	Very Low	Very High
Output Impedance	Medium-High	Very High	Very Low
Common Use	General purpose ★	High freq / RF amplifiers	Impedance matching / Emitter follower



§ 4.1



BLOCK	FUNCTION	EXAMPLES
Sensor	Detects the physical quantity to be measured	Thermocouple, Strain gauge, Photodiode
Transducer	Converts non-electrical quantity → electrical signal (voltage/current)	Piezoelectric, LVDT, RTD
Signal Conditioning	Amplifies, filters, or normalizes the transducer output for processing	Op-Amp, Filter, Bridge circuit
A/D Converter	Converts analog signal to digital for processing/storage	ADC chips, Microcontroller ADC
Display / Recorder	Shows measured value to user or stores data	LCD/LED meter, Oscilloscope, Data logger



§ 4.2

Sensor

A device that responds to a physical stimulus (heat, light, pressure, sound) and produces a signal. Every transducer contains a sensor.

Transducer

A device that converts one form of energy to another, specifically converting non-electrical quantities into electrical signals for measurement.

MEASURAND	TRANSDUCER USED	OUTPUT
Temperature	Thermocouple / RTD	mV / Resistance change
Pressure	Piezoelectric / Strain gauge	Voltage / Resistance change
Displacement	LVDT (Linear Variable Differential Transformer)	Voltage
Light Intensity	Photodiode / LDR	Current / Resistance change
Sound	Microphone	Voltage (AC)



§ 4.3

Accuracy

Closeness of a measured value to the true value. Expressed as $\pm \%$.

Precision

Reproducibility of repeated measurements.
High precision $\neq$ high accuracy.

Sensitivity

Ratio of change in output to change in input: $S = \Delta \text{Output} / \Delta \text{Input}$.

Resolution

Smallest change in input detectable by the instrument.

Range

The span of values the instrument can measure (min to max).

Error

Difference between measured and true value:
 $\text{Error} = \text{Measured} - \text{True}$.



§ 4.4

PARAMETER	NORMAL PN DIODE	ZENER DIODE
Normal Operation	Forward biased (conducts at $\sim 0.7V$ Si)	Reverse biased (breakdown region)
Breakdown	Damages the diode	Designed to operate at breakdown
Voltage Regulation	Not possible	Maintains constant V_Z
Doping Level	Lightly doped	Heavily doped (sharp breakdown)
Application	Rectification, switching	Voltage regulation, reference, protection

Symbol	Standard diode arrow	Diode with bent cathode line (Z-shape)
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§ 5.1

Q1. What is a rectifier?

A device that converts AC (alternating current) to DC (direct current / pulsating DC) using diodes.

Q3. What is the ripple factor?

$\gamma = V_{ac}/V_{dc}$. Ratio of AC component to DC component. Lower is better. HWR: 1.21; FWBR: 0.48.

Q5. How many diodes does a bridge rectifier use?

Four (4) diodes arranged in a bridge configuration. No center-tap transformer required.

Q7. What is the working region of a Zener diode?

Reverse breakdown region. It maintains constant voltage V_Z across its terminals in this region.

Q9. What does β (beta) represent?

Current gain of a BJT transistor: $\beta = I_C/I_B$. Also written as h_{fe} . Typical range: 50–300.

Q11. What is the –3 dB point?

The frequency at which gain drops to $1/\sqrt{2}$ (0.707) of its maximum value. Defines bandwidth edges.

Q13. What is a transducer?

A device that converts one form of energy (non-electrical) into another (electrical) for measurement purposes.

Q15. What is V_{dc} for FWBR?

$V_{dc} = 2V_m/\pi \approx 0.636 V_m$, where V_m is the peak output voltage of the transformer secondary.

Q2. What is ripple in rectifier output?

The residual AC component present in the rectified (pulsating DC) output. It causes unwanted fluctuations.

Q4. What is the function of a capacitor filter?

Reduces ripple by charging to peak voltage during diode conduction and discharging through R_L during off period.

Q6. What is efficiency of FWBR?

81.2%. It uses both half cycles of AC input, making it more efficient than HWR (40.6%).

Q8. What is the phase shift in CE amplifier?

180° — the output is inverted (out of phase) with respect to the input signal.

Q10. What is bandwidth of an amplifier?

$BW = f_H - f_L$. The range of frequencies for which the amplifier gain is at least 70.7% of max gain.

Q12. What is the function of coupling capacitors in CE amplifier?

C_1 and C_2 block DC and allow only AC signals to pass between stages, preventing bias disturbance.

Q14. What is signal conditioning?

The process of amplifying, filtering, or converting transducer output to a form suitable for display or processing.



§ 5.2

Q16. What is the role of transformer in DC power supply?

Steps down mains voltage (230V AC) to required lower AC voltage and provides isolation from mains.

Q18. What is the purpose of R_E in CE amplifier?

Provides DC stability (thermal stability). Bypassed by C_E for AC to avoid gain reduction.

Q20. Why is CE configuration most commonly used?

It provides the highest power gain among all BJT configurations, with both voltage and current gain > 1 .

Q22. What causes gain to fall at low frequencies?

Coupling and bypass capacitors have high impedance at low frequencies, reducing signal transmission and gain.

Q17. What is the output frequency of FWBR?

$2f = 100$ Hz (for 50 Hz mains supply). Both half cycles appear at output.

Q19. What is power gain formula?

$A_p = A_v \times A_i$. In dB: $A_p(\text{dB}) = 10 \log(P_{out}/P_{in})$.

Q21. What causes gain to fall at high frequencies?

Internal junction capacitances of the BJT short-circuit at high frequencies, reducing the effective impedance and gain.

Q23. Define accuracy in instrumentation.

Closeness of the measured value to the true (actual) value. Expressed as $\pm$ percentage of full scale.

Q24. What are diodes D1, D3 in FWBR doing in +ve half cycle?

D1 and D3 conduct, D2 and D4 are reverse biased. Current flows through load in one direction.

Q25. What is the formula for sensitivity of a transducer?

$S = \Delta \text{Output} / \Delta \text{Input}$. Higher sensitivity means smaller input changes produce larger detectable output changes.



§ 6.1

⚠ COMMON ERRORS TO AVOID

- ⚠ Forgetting 180° phase reversal in CE amplifier — Always mention it!
- ⚠ Confusing HWR and FWBR formulas — HWR: $0.318V_m$; FWBR: $0.636V_m$
- ⚠ Wrong ripple factor comparison — Lower γ = better; FWBR is better
- ⚠ Missing units in formulas — Volts, Amps, Hertz, Farads
- ⚠ Zener vs normal diode confusion — Zener works in reverse breakdown!
- ⚠ Skipping block diagram explanations — Always explain each block
- ⚠ Forgetting negative sign in A_v — $A_v = -\beta R_C / r_e$
- ⚠ Misunderstanding ripple factor formula — $\gamma = V_{ac}/V_{dc}$, NOT V_{ac}/V_{rms}
- ⚠ Wrong dB formulas — Voltage: 20log; Power: 10log
- ⚠ Confusing bandwidth definition — $BW = f_H - f_L$, not just f_H

RECTIFIER TIPS

- ▶ FWBR has **4 diodes** in bridge, NO center tap needed
- ▶ Center-tap FWR uses **2 diodes** but needs CT transformer
- ▶ Ripple factor: lower = smoother DC (FWBR wins)
- ▶ Capacitor filter: larger C → smaller ripple → better
- ▶ PIV (Peak Inverse Voltage) = V_m for bridge rectifier
- ▶ Efficiency: FWBR = 81.2% (double of HWR's 40.6%)

ZENER REGULATOR TIPS

- ▶ Zener in **reverse bias** — key point!
- ▶ Series resistor R_s is essential — limits current
- ▶ Output = V_Z (zener breakdown voltage) — constant
- ▶ $I_s = I_Z + I_L$ — current divides between Zener and load
- ▶ When load decreases → I_Z increases to compensate

AMPLIFIER TIPS

- ▶ **Phase reversal = 180°** in CE — NEVER forget
- ▶ $\beta = h_{fe}$ (same thing, different notation)
- ▶ $A_v(\text{dB}) = 20 \log |A_v|$; $A_p(\text{dB}) = 10 \log A_p$
- ▶ Midband = flat gain region = useful operating range
- ▶ At -3 dB points: gain = $0.707 \times A_{\text{max}}$
- ▶ Bypass cap C_E shorts R_E → increases gain

INSTRUMENTATION TIPS

- ▶ All transducers are sensors, but not vice versa
- ▶ Signal conditioning always comes after transducer
- ▶ Accuracy ≠ Precision — understand the difference!
- ▶ Block diagram order: Sensor → Conditioning → Display
- ▶ ADC converts analog signal to digital for processing



§ 6.2

QUANTITY	HWR	FWBR
V_{dc}	$0.318 V_m$	$0.636 V_m$
V_{rms}	$V_m/2 = 0.5 V_m$	$V_m/\sqrt{2} = 0.707 V_m$
Ripple Factor (γ)	1.21	0.48
Efficiency (η)	40.6%	81.2%
Form Factor (FF)	1.57	1.11
Output Frequency	f (50 Hz)	$2f$ (100 Hz)
No. of Diodes	1	4



⚡ RECTIFIER FORMULAS

HWR V_{DC}

$$V_{dc} = V_m / \pi$$

$$= 0.318 V_m$$

FWBR V_{DC}

$$V_{dc} = 2V_m / \pi$$

$$= 0.636 V_m$$

 V_{RMS} (FWBR)

$$V_{rms} = V_m / \sqrt{2}$$

$$= 0.707 V_m$$

RIPPLE FACTOR

$$\gamma = V_{ac} / V_{dc}$$

$$\text{HWR:1.21} \mid \text{FWBR:0.48}$$

EFFICIENCY

$$\eta = P_{dc} / P_{ac}$$

$$\text{HWR:40.6\%} \mid \text{FWBR:81.2\%}$$

FORM FACTOR

$$FF = V_{rms} / V_{dc}$$

$$\text{HWR:1.57} \mid \text{FWBR:1.11}$$

🔧 FILTER FORMULAS

RIPPLE VOLTAGE

$$V_r = I_{dc} / (f \cdot C)$$

$$f = \text{frequency, } C = \text{capacitance}$$

RIPPLE FACTOR (WITH CAP FILTER)

$$\gamma = 1 / (4\sqrt{3} \cdot f \cdot C \cdot R_L)$$

$$\text{Larger } C \rightarrow \text{smaller } \gamma$$

⚡ ZENER REGULATOR FORMULAS

SERIES RESISTOR CURRENT

$$I_s = (V_{in} - V_Z) / R_s$$

CURRENT DIVIDES

$$I_s = I_Z + I_L$$

$$\text{Between Zener \& Load}$$

REGULATED OUTPUT

$$V_{out} = V_Z$$

$$\text{Constant breakdown voltage}$$

🔊 CE AMPLIFIER FORMULAS

VOLTAGE GAIN

$$A_v = -\beta \cdot R_C / r_e$$

$$\text{—ve} = 180^\circ \text{ phase shift}$$

CURRENT GAIN

$$A_i = \beta = h_{fe}$$

$$I_C / I_B; \text{range } 50\text{--}300$$

POWER GAIN

$$A_p = A_v \times A_i$$

$$\text{In dB: } 10 \log(P_{out} / P_{in})$$

VOLTAGE GAIN (DB)

$$A_v(\text{dB}) = 20 \log|A_v|$$

BANDWIDTH

$$BW = f_H - f_L$$

$$\text{—3dB} = 0.707 \times A_{max}$$

DYNAMIC EMITTER RESISTANCE

$$r_e = 26\text{mV} / I_E$$

$$\text{At room temperature}$$

🔧 INSTRUMENTATION

SENSITIVITY

$$S = \Delta \text{Output} / \Delta \text{Input}$$

$$\text{Higher } S = \text{more sensitive}$$

PERCENTAGE ERROR

$$\text{Error\%} = [(M - T) / T] \times 100$$

$$M = \text{Measured, } T = \text{True value}$$



§ 8.1

Rectifier

Circuit that converts AC $\rightarrow$ pulsating DC using diodes. Types: HWR, FWBR.

Ripple

AC component remaining in rectified output. Reduced by filters.

Filter

Circuit that smooths pulsating DC. Capacitor (C) placed parallel to load.

Regulator

Maintains constant output voltage despite load or input changes.

Zener Diode

PN junction designed to operate in reverse breakdown for voltage regulation.

Amplifier

Circuit that increases amplitude of input signal without changing its frequency.

CE Configuration

Emitter common to input & output. Gives max power gain with 180° phase shift.

Bandwidth

Frequency range between lower (f_L) and upper (f_H) -3dB cutoff points.

Transducer

Converts non-electrical quantity into an electrical signal for measurement.

Signal Conditioning

Amplifying, filtering, or scaling of transducer output for display/processing.

Sensitivity

$\Delta\text{Output}/\Delta\text{Input}$ — ability of instrument to detect small changes in measurand.

PIV

Peak Inverse Voltage — max reverse voltage a diode must withstand. FWBR: V_m .



§ 8.2

RECTIFIERS:AC $\rightarrow$ DC

Pulsating DC

4 Diodes (Bridge)

0.636 V_m $\gamma=0.48$ $\eta=81.2\%$

100Hz output

PIV= V_m **FILTER & REGULATOR:**

Parallel Capacitor

Charge-Discharge

Ripple Reduction

Zener Breakdown

Reverse Bias

Constant V_{out} Series R_s **CE AMPLIFIER:** 180° Phase Shift β (beta)

High Power Gain

Coupling Cap

Bypass Cap

 -3dB point

Midband

 $r_e=26\text{mV}/I_E$ **INSTRUMENTATION:**Sensor $\rightarrow$ Transducer

Signal Conditioning

ADC

Display

Accuracy

Precision

Resolution

Sensitivity



§ 8.3

① DC Power Supply

AC Input



Transformer



Rectifier



Filter



Regulator



DC Output

② PA System

Microphone
(Sound $\rightarrow$ Signal)Pre-Amplifier
(Boost)Power Amplifier
(Drive)Loudspeaker
(Signal $\rightarrow$ Sound)

③ Instrumentation System

Physical
QuantitySensor/
TransducerSignal
Conditioning

ADC

Display/
Record**All the Best for Your Exam!** Remember: Phase reversal = 180° | FWBR efficiency = 81.2% | Zener = Reverse breakdown | $BW = f_H - f_L$

